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Paper 1: Early exposure-response modeling of an interferon-beta monoclonal antibody (dazu)

Bibliographic Information

- **Title:** Early exposure-response modeling of an interferon-beta monoclonal antibody (dazukibart) in adults with dermatomyositis
- **Authors:** Prybylski JP, Zhu J, Banfield C, et al.
- **Journal:** Journal of Pharmacokinetics and Pharmacodynamics, 2026, 53:17
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Paper Type: POPK_PKPD

Quick Take

This paper describes a joint exposure-response model for dazukibart in dermatomyositis (DM), integrating multiple clinical endpoints to predict Total Improvement Score (TIS) in skin-predominant patients where TIS was not directly measured. Pharmacometricians working on rare diseases with sparse, multi-endpoint data will find the modeling approach instructive. The model supports dose selection and trial design for Phase 3 development.

Executive Summary

Prybylski et al. developed a comprehensive exposure-response model linking dazukibart pharmacokinetics and interferon-beta (IFN β) binding to multiple clinical outcomes in DM, including CDASI, SF-36, global assessments, and TIS. The model uses a shared effect driven by free IFN β fraction, with endpoint-specific maximum effects and a placebo component, all within an indirect response framework. By leveraging correlations among endpoints, the model predicts TIS response in skin-predominant DM patients, a critical endpoint not measured in early trial stages. The approach demonstrates how joint modeling of multiple outcomes can compensate for limited data in rare diseases, providing a robust basis for dose selection and trial simulations. The model's predictions were consistent with external trial data, supporting its utility for model-based meta-analysis and future DM drug development.

Scientific Context & Motivation

Dermatomyositis is a rare autoimmune disease with both skin and muscle manifestations, making clinical trial design challenging due to small patient numbers and heterogeneous presentation. The Total Improvement Score (TIS) is a holistic endpoint combining muscle, skin, and global assessments, but it was only measured in a small subset of muscle-predominant patients in the Phase 2 study of dazukibart. To inform continued development, there was a need to predict TIS response in skin-predominant patients and to understand the exposure-response relationship across all relevant endpoints. This analysis fills the gap by building a joint model that shares

information across endpoints, enabling extrapolation to unmeasured outcomes and supporting dose selection for future trials.

✂ Methodological Snapshot

The analysis used a sequential PK/PD-to-ER approach. A two-compartment PK model with first-order elimination described dazukibart concentrations. IFN β binding was modeled via a quasi-steady-state approximation, yielding individual predictions of free IFN β fraction. The ER model assumed a shared effect proportional to free IFN β , driving multiple clinical endpoints through indirect response models (Type II/IV). Endpoint-specific E_{max} parameters captured differential sensitivity, and a placebo effect component was included. Inter-individual variability was modeled with a block covariance matrix, and shared random effects were used to handle high correlations. Model fitting was performed in NONMEM with FOCE-I; uncertainty was assessed via SIR.

☒ Structural Model Breakdown

The ER model consists of a series of indirect response models for each clinical endpoint m , driven by a shared effect E_{shared} . The shared effect is computed as $E_{shared} = E_{max,norm} \cdot P_{unbound}$, where $P_{unbound} = 1 - C/(C + K_{SS})$ is the fraction of free IFN β . For each endpoint, the dynamics are given by $dEnd_m/dt = k_{in,m} - k_{out} \cdot End_m \cdot (1 + E_{max,m} \cdot E_{shared})$, with $k_{in,m} = k_{out} \cdot End_{base,m}$. The baseline $End_{base,m}$ is a corrected value accounting for the maximal IFN β effect at baseline: $End_{base,m} = End_{base,DM,m} \cdot (1 + E_{max,m} \cdot E_{max,norm}) \cdot E_{plac}^{sign(E_{max,m})}$. The initial condition is $End_m(0) = End_{base,DM,m}$. Placebo effect is incorporated via E_{plac} , which shifts the steady-state. IIV is included on baselines (modified log-normal), E_{plac} (log-normal), and $E_{max,norm}$ (logit-normal). Correlations among baseline IIVs are captured by a full block ω matrix; for highly correlated endpoints, random effects are shared via linear combinations with fixed-effect scalars. Residual error is additive on the identity scale, except for muscle enzymes which are modeled as ratio to ULN.

Detailed Methodological Analysis

Modeling Approach

The exposure-response model linked dazukibart exposure to clinical outcomes via IFN β binding. The PK model was a standard two-compartment mAb model. The PD model used a quasi-steady-state approximation to relate dazukibart concentration to bound IFN β . The ER model assumed that the fraction of unbound IFN β ($P_{unbound}$) drives a shared effect ($E_{shared} = E_{max,norm} \cdot P_{unbound}$), which then modulates each clinical endpoint through an indirect response model. Each endpoint m had its own baseline ($End_{base,DM,m}$), maximum effect ($E_{max,m}$), and was influenced by a common turnover rate (k_{out}) and placebo effect (E_{plac}). The model was implemented as a set of differential equations with initial conditions at the observed DM baseline, which was treated as an altered steady-state due to pre-existing IFN β effect. Covariance among baseline IIVs was

estimated using a block ω matrix; shared etas were used to reduce dimensionality. The model was developed in NONMEM 7.5.0 using FOCE-I and ADVAN13.

Data Sources

Data came from the Phase 2 study C0251002, an adaptive, randomized, double-blind, placebo-controlled trial in adults with DM. The study had multiple stages: Stage 1 (SPDM, 600 mg vs placebo), Stage 2 (SPDM, 150 mg and 600 mg vs placebo), Amended Stage 2 (crossover), and Stage 3 (MPDM, 600 mg vs placebo). Endpoints included CDASI-A, CDASI-D, SF-36 MCS and PFD, PhGA, PtGA, and TIS subcomponents (MMT-8, HAQ, muscle enzymes, ExGA). TIS was only measured in Stage 3 (MPDM). Demographics are summarized in Table 1.

Estimation Methods

Model parameters were estimated using First-Order Conditional Estimation with Interaction (FOCE-I) in NONMEM 7.5.0. The ADVAN13 subroutine was used for differential equations. SAEM was explored but did not improve stability. Parameter uncertainty was assessed using Sampling Importance Resampling (SIR) via PsN, with 1000 samples and 500 resamples. A limited bootstrap ($N = 200$, stratified by stage) was performed to evaluate base model stability.

Model Evaluation

Model evaluation included standard diagnostic plots (observed vs predicted, conditional weighted residuals vs time/predictions), visual predictive checks (VPCs) stratified by relevant covariates, and prediction-corrected VPCs for TIS categories. The model's predictive performance for TIS in SPDM was informally compared with external data from IVIG and lenabasum trials. Shrinkage was assessed for all random effects.

Covariate Analysis

Due to limited sample size, only the effect of MPDM status (muscle-predominant vs skin-predominant) was tested on shared structural parameters and baseline estimates. Covariate selection used a stepwise forward addition ($\alpha = 0.01$) and backward elimination ($\alpha = 0.001$). Final covariate effects included adjustments on CDASI-A and SF-36 PFD baselines and on E_{plac} for MPDM patients.

Statistical Rigor Assessment

The statistical methods are appropriate for the complexity of the data. FOCE-I is a standard estimation method for population PK/PD models. The use of SIR for uncertainty is robust, especially given the high number of parameters. The bootstrap success rate of 72% indicates some model instability, likely due to the large covariance matrix and limited data in some endpoints. The authors acknowledge this and note that failures were due to rounding errors. Shrinkage was acceptable ($< 25\%$). The covariate analysis, though limited, followed a rigorous stepwise procedure. The main statistical limitation is the assumption of continuous normal residuals for bounded/discrete endpoints, which could affect parameter estimates and uncertainty. Additionally, the sequential IPP approach may underestimate uncertainty in ER parameters. Overall, the

analysis is statistically sound for its exploratory purpose, but confirmatory trials would benefit from a fully simultaneous Bayesian approach.

☒ Key Findings

The model successfully described the time courses of CDASI, SF-36, PhGA, PtGA, and TIS sub-components using a shared effect driven by the fraction of unbound IFN β . All unfixed endpoint-specific E_{max} parameters excluded zero, suggesting dazukibart efficacy across multiple domains. The placebo effect parameter (E_{plac}) was not significantly different from 1, indicating minimal placebo response in typical skin-predominant DM patients. Clinical trial simulations predicted that 600 mg every 4 weeks would yield greater TIS response than placebo in both skin- and muscle-predominant patients, with major response rates increasing over time. External validation against IVIG and lenabasum trial data showed consistency with model predictions, supporting the model's credibility.

🔗 Clinical & Regulatory Implications

The model supports the continued development of dazukibart by providing evidence of TIS improvement in both SPDM and MPDM patients. The predicted TIS response rates at 600 mg every 4 weeks can inform Phase 3 trial design, including sample size calculations and endpoint selection. The finding of minimal placebo effect in SPDM suggests that placebo-controlled trials may be feasible without large placebo responses confounding results. The model also indicates that dazukibart may offer benefits across multiple disease domains, which is important for a holistic treatment approach. Regulatory agencies may view the joint modeling approach favorably as a way to leverage limited data in rare diseases, though they would likely require prospective confirmation of TIS predictions in SPDM patients.

Strengths & Limitations

Strengths

- Innovative joint modeling approach that shares information across multiple endpoints to compensate for sparse data.
- Semi-mechanistic link from drug exposure to IFN β binding to clinical outcomes, providing a pharmacological rationale.
- Use of a shared effect parameter (E_{shared}) and endpoint-specific E_{max} allows flexible description of differential response.
- Incorporation of placebo effect with directional consistency.
- Thorough model evaluation including SIR for parameter uncertainty, VPCs, and external validation against independent trials.
- Application of item response theory concepts to pharmacometrics.

Limitations (Acknowledged by Authors)

- Shared parameters (k_{out} , $E_{max,norm}$, E_{plac}) may not fully capture endpoint-specific dynamics; some endpoints may have different placebo effects.
- Assumption that TIS in skin-predominant patients can be predicted without direct observation is strong and may not hold if correlations differ.
- Numerical properties of bounded/discrete endpoints were ignored, potentially biasing parameter estimates (e.g., k_{out}).
- Limited covariate analysis due to small sample size; only MPDM status was tested.
- Sequential IPP approach may introduce bias compared to simultaneous PK/PD-ER fitting.

Limitations (Expert Review)

- The model complexity (many random effects, shared parameters) may lead to overfitting or instability; bootstrap success rate was 72%.
- The use of a single k_{out} for all endpoints assumes identical time-to-equilibrium, which is unlikely for diverse clinical scales.
- The model does not account for potential disease progression or time-dependent changes in placebo response beyond the simple E_{plac} factor.
- External validation was ad hoc and informal; a formal meta-analysis would strengthen conclusions.
- The model code provided is redacted, limiting full reproducibility.

Generalizability

The modeling framework is generalizable to other rare diseases with multiple endpoints and limited data, particularly where a pharmacological marker can serve as a driver. However, the specific parameter estimates and predictions are tied to dazukibart and the DM population studied; extrapolation to other drugs or diseases would require careful adaptation.

Key Equations

IFN β total mass balance

$$IFN_{total} = IFN_{unbound} + IFN_{bound}$$

Defines the relationship between total, unbound, and bound interferon-beta.

Fraction of unbound IFN β

$$P_{unbound} = \frac{IFN_{unbound}}{IFN_{total}} = 1 - \frac{IFN_{bound}}{IFN_{total}} = 1 - \frac{C}{C + K_{SS}}$$

Shows how the fraction of free IFN β depends on dazukibart concentration C and the quasi-steady-state binding constant KSS.

Shared effect

$$E_{shared} = E_{max, norm} \cdot P_{unbound}$$

The shared pharmacological effect is proportional to the fraction of unbound IFN β , scaled by a normalizing maximum effect parameter.

Corrected baseline for endpoint m

$$End_{base,m} = End_{base, DM,m} \cdot (1 + E_{max,m} \cdot E_{max, norm}) \cdot E_{plac}^{\text{sign}(E_{max,m})}$$

Calculates the IFN β - and placebo-corrected baseline used to derive the input rate; accounts for the pre-existing maximal IFN β effect at baseline.

Input rate for endpoint m

$$k_{in,m} = k_{out} \cdot End_{base,m}$$

The zero-order production rate of the clinical endpoint is set to maintain the corrected baseline at steady state.

Indirect response model for endpoint m

$$\frac{dEnd_m}{dt} = k_{in,m} - k_{out} \cdot End_m \cdot (1 + E_{max,m} \cdot E_{shared})$$

Describes the dynamics of each clinical endpoint as a Type II/IV indirect response model, where the shared effect modulates the loss rate.

Shared inter-individual variability

$$\eta_{1i}^* = \theta_{X1,1} \cdot \eta_{1i} + \theta_{X1,2} \cdot \eta_{2i} + \theta_{X1,3} \cdot \eta_{3i} + \dots$$

When random effects are highly correlated, an individual's eta for one endpoint can be expressed as a linear combination of other etas, reducing the number of estimated variance parameters.

Enzyme observation model

$$Y_{ij, enz} = (R_{ij, enz} + \varepsilon_{ij, enz}) \cdot ULN_{enz}$$

For muscle enzymes, the observed value is modeled as the sum of the predicted ratio to ULN and additive error, then scaled back to the original units.

Example of shared IIV for ExGA

$$\eta_{ExGA}^* = \eta_{ExGA} + \theta_{X PhGA, ExGA} \cdot \eta_{PhGA}$$

Demonstrates the application of shared etas: the random effect for Extramuscular Global Assessment is partly informed by the PhGA random effect.

Figures & Tables

- **Figure 1:** Diagram of the exposure-response model structure, showing the relationship between dazukibart PK, IFN β binding, shared effect, and individual endpoints.

- *Significance*: Provides a visual overview of the model’s mechanistic and statistical components.
 - **Figure 2**: Diagnostic plots including observed vs predicted, conditional weighted residuals, and VPCs for key endpoints.
 - *Significance*: Demonstrates model adequacy and predictive performance.
 - **Figure 3**: Visual predictive check for TIS continuous score and responder categories.
 - *Significance*: Shows the model’s ability to predict TIS, the primary endpoint of interest, including in skin-predominant patients.
 - **Figure 4**: Time course of CDASI-A response stratified by DM type (SPDM vs MPDM).
 - *Significance*: Illustrates differential response and model fit in the two subpopulations.
 - **Figure 5**: Clinical trial simulation results showing predicted TIS response rates for placebo and 600 mg dazukibart in mixed DM populations.
 - *Significance*: Supports dose selection and trial design by projecting treatment effects.
 - **Table 1**: Demographics and baseline outcomes data by study stage.
 - *Significance*: Describes the study population and highlights differences between skin- and muscle-predominant groups.
 - **Table 2**: Final model parameter estimates with SIR-derived confidence intervals.
 - *Significance*: Provides the quantitative basis for the model and shows the precision of key parameters.
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Code & Reproducibility Assessment

Partial model code (with proprietary details removed) is provided in Supplementary Materials. Full code and data are available upon request from Pfizer, subject to review.

Supplementary Materials

Supplementary materials include additional model details, parameter tables (e.g., base model estimates, covariance parameters), diagnostic figures for other endpoints, and the redacted NONMEM model code.

Future Directions

Future work could refine the model by incorporating bounded integer or beta regression to respect endpoint distributions, and by using a fully Bayesian framework to handle the complex covariance structure. A formal model-based meta-analysis combining dazukibart data with other DM trials would strengthen the evidence for TIS predictions. The approach could be extended to include latent variable models (e.g., IRT) for composite scores, and to explore the impact of different dosing regimens and patient subtypes.

Expert Commentary

As a senior pharmacometrician, I find this paper a valuable contribution to the field of rare disease drug development. The authors tackle a common problem: how to make decisions when

key endpoints are missing in a substantial portion of the target population. Their solution—building a joint model that borrows strength across endpoints—is elegant and pragmatic. The use of a pharmacodynamic marker (free IFN β) as the driver, rather than a purely latent variable, adds mechanistic credibility. However, the model’s complexity demands careful diagnostics; the 72% bootstrap success rate suggests some fragility. I would have liked to see a sensitivity analysis comparing the joint model to simpler alternatives, and a more rigorous treatment of the bounded nature of the scores. Nevertheless, the paper provides a template for others facing similar challenges, and the external consistency checks are reassuring. The discussion of IRT and latent variable models places the work in a broader methodological context, which is helpful for readers. Overall, this is a well-executed analysis that advances the practice of model-informed drug development in rare diseases.

Bottom Line

This work exemplifies how joint exposure-response modeling of multiple clinical endpoints can extract maximal information from sparse rare-disease trials, enabling prediction of unmeasured outcomes and informed dose selection. Practitioners should consider similar approaches when faced with fragmented efficacy data, but must be mindful of the strong assumptions regarding shared parameters and endpoint correlations.

☒ Figures

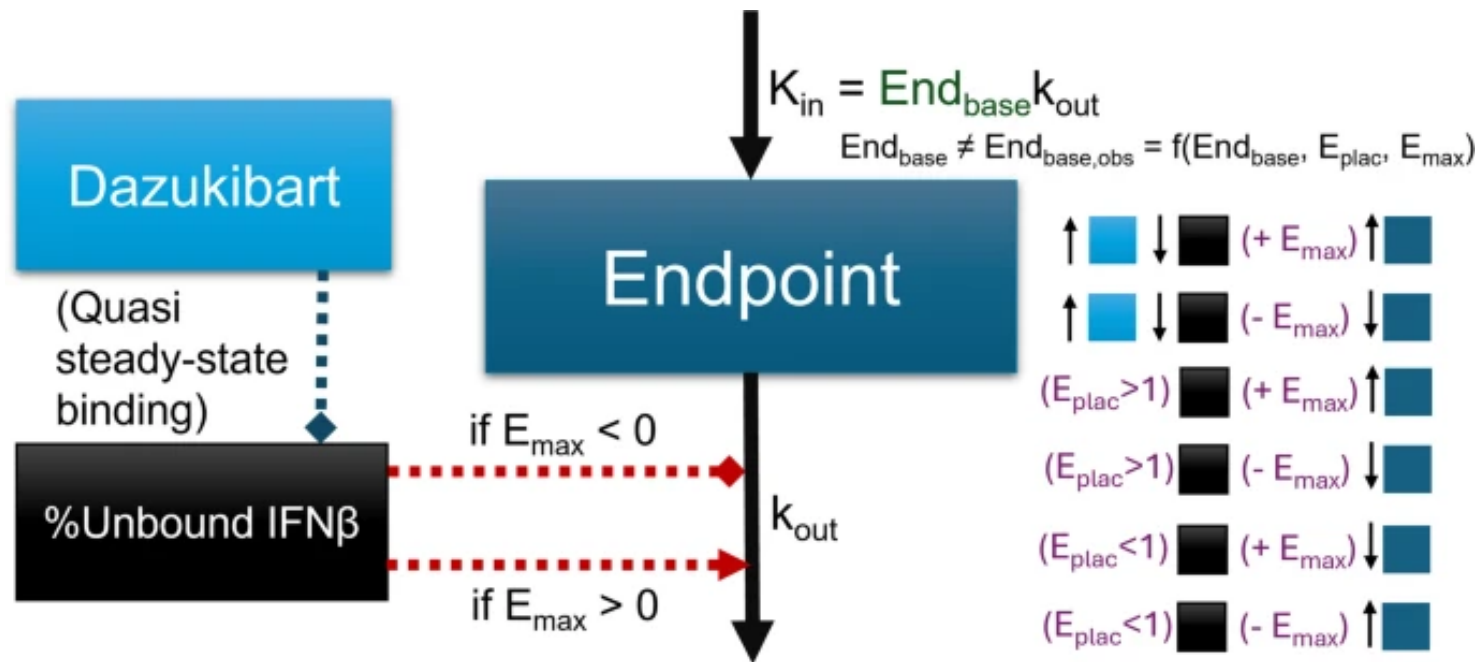


Figure 1: Figure 1

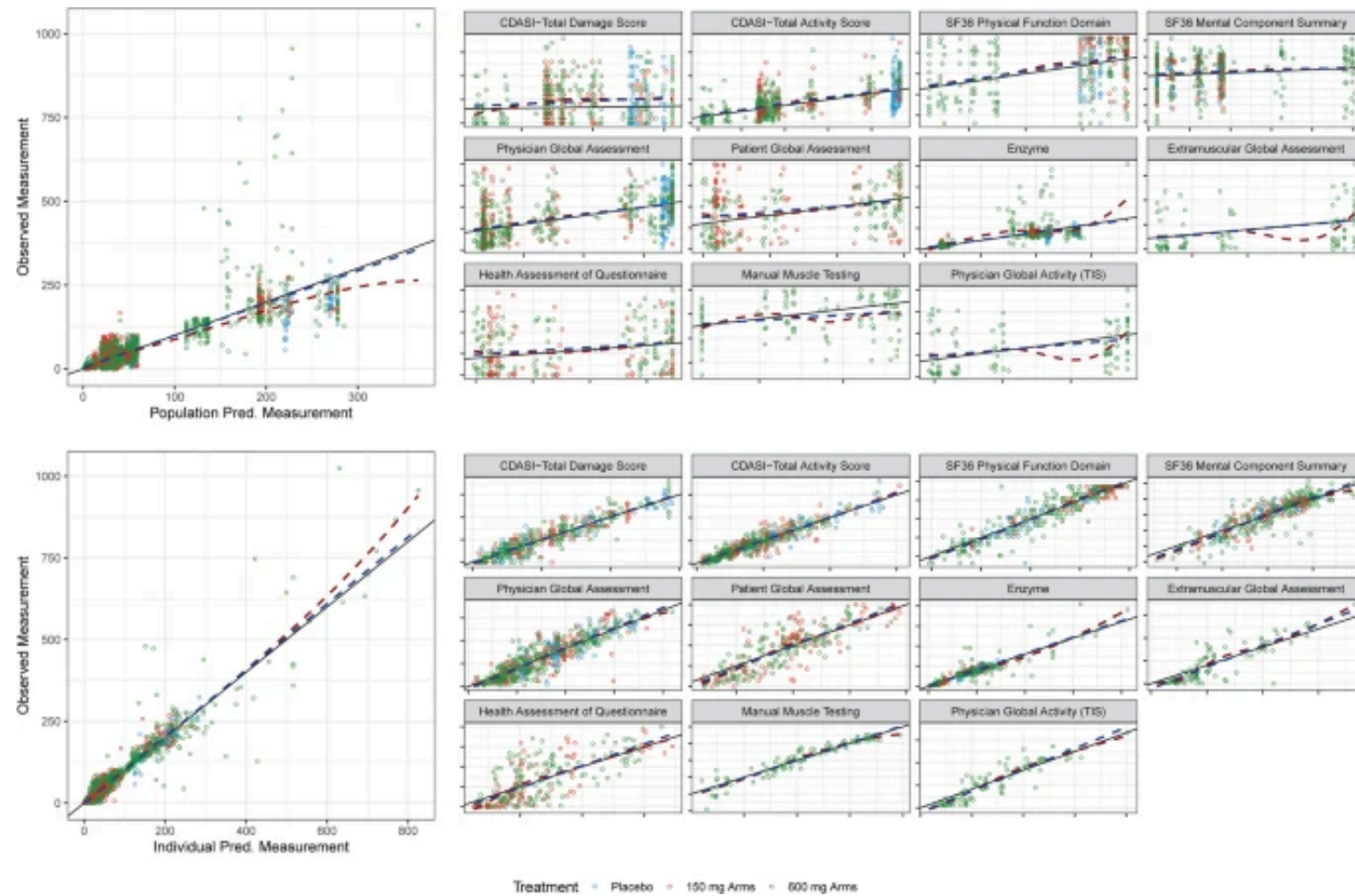


Figure 2: Figure 2

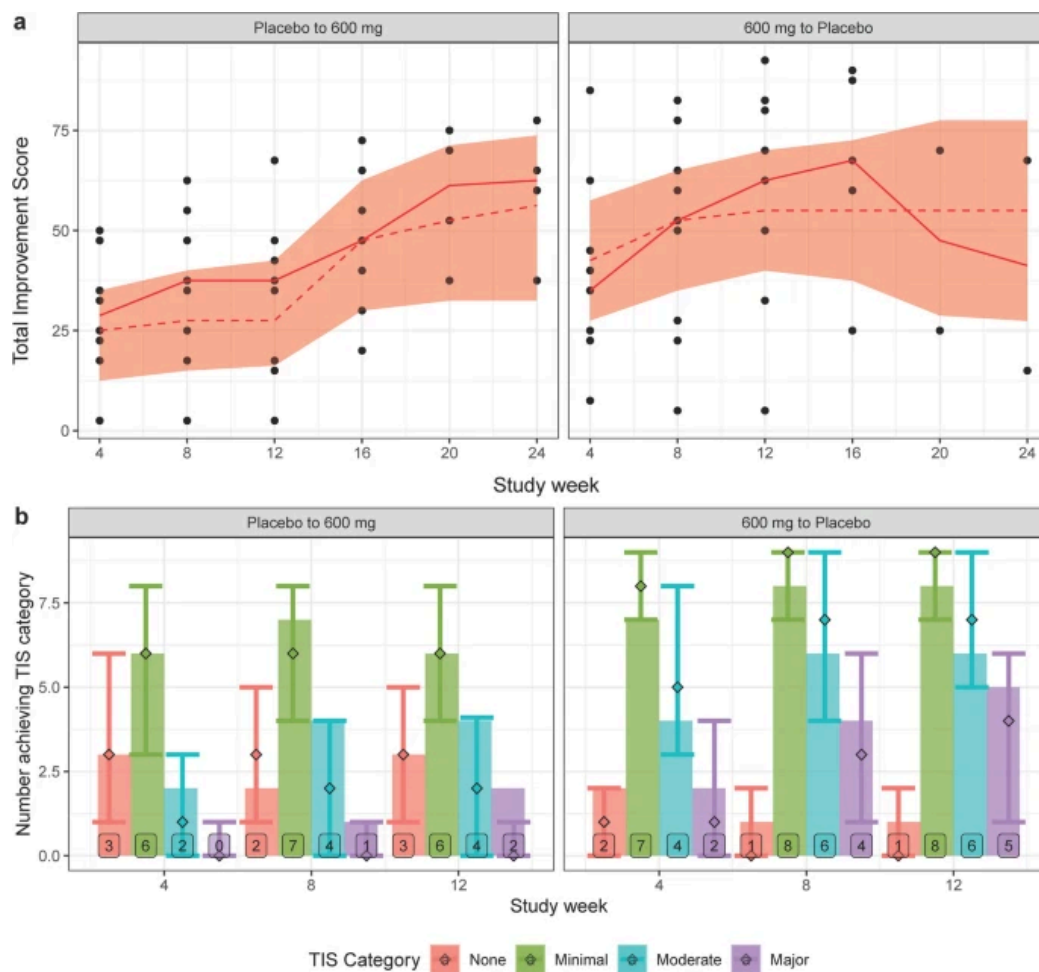


Figure 3: Figure 3

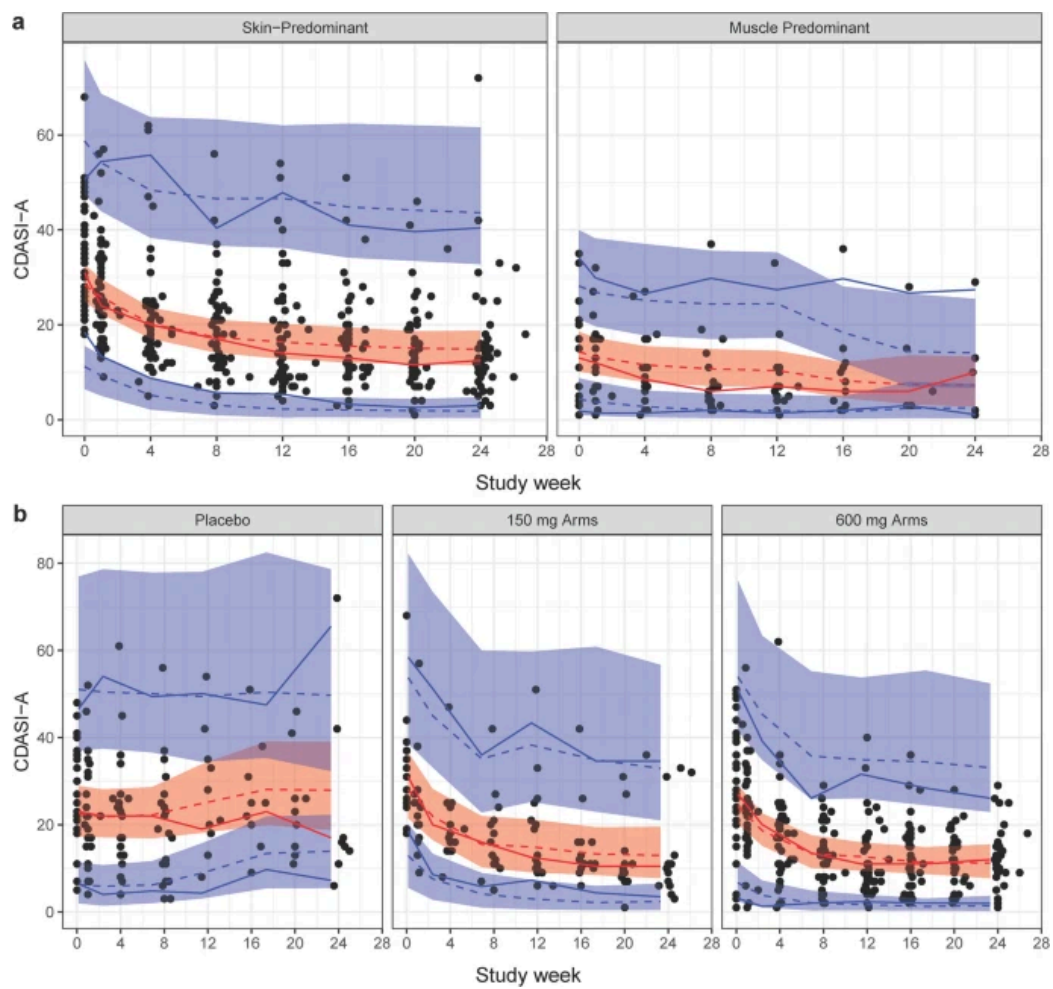


Figure 4: Figure 4

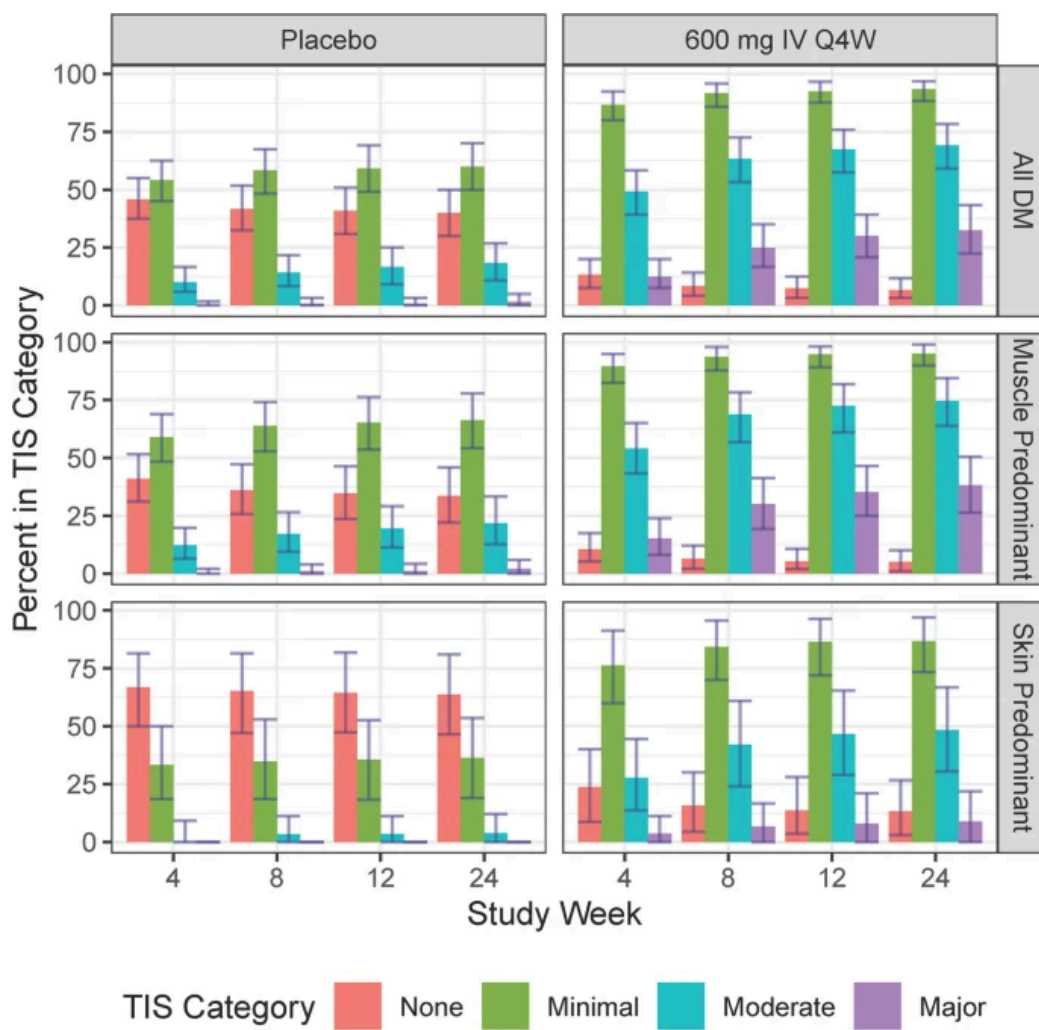


Figure 5: Figure 5